Module 3 – Characterization of a Nonlinear Spring

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As the program manager of the computing division, Anthony often supervises computational engineers by assigning the development of numerical models, ensuring task alignment with project goals, and delivering client-facing work. He communicates complex technical work into clear insights for clients, helping them understand the implications of the models and simulations, thus guiding them make informed decisions about the project's direction. Anthony needs to highlight key findings, potential risks, and proposed solutions, emphasizing how the computational work supports the client's objectives. He facilitates communication between the engineering team and clients, ensuring that feedback is incorporated and the project stays on track.

Characterization of a Nonlinear Spring for Webb Motorsports

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From: anthony.jordan@burdell.com
To: you.employee@burdell.com

Dear [Computational Engineer]:

Because of your success in the last two projects, Webb Industries has come to us again, this time to support their motorsports division. They are designing a strut suspension with a new type of coil spring that they have been developing for some time. It has nonlinear behavior in that it hardens as the spring is extended or compressed, and they would like our help in characterizing the most recent iteration of this spring they developed in their production facilities.Please see below for the details.Furthermore – up to this point I have been providing you with specific guidance about what to include in your presentation documents. At this point I would like to give you a little more flexibility.

Project Overview:
A spring is one of the simplest elements in most mechanical models because it accounts for forces acting through the spring as a simple ratio of displacement:

$$F = kx$$

Where is the spring force in newtons $[N]$, x is the spring displacement in meters $[m]$, k is the spring, or Hooke's, constant in units $[N/m]$. This assumes a spring is acting linearly, but the current spring from Webb Motorsports is specifically designed to act nonlinearly. They believe that it offers

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superior racing performance and would like to characterize it so they can manufacture more of them. The problem that the client is having is that they don't know what type of nonlinearity this spring is governed by.The client would like us to investigate this issue with some data they have collected from real-world driving tests as well as some precisely obtained laboratory data.

Tasks:

1. **Regression Modeling Analysis**
   - **Webb Motorsports is requesting help in determining which model best describes the spring they have installed in their vehicle.** They have provided us with a .mat file that contains spring displacement data and force sensor data for the spring during a driving simulation called `drivetest.mat` . Please choose 6 different types of fits that make some sort of physical or modeling sense, one of which is a linear fit, and compare them to find the **best** resulting model. Please verify that these models are appropriate to use in later statistical analyses. Report the coefficients and governing equation for your resulting best model compared to the linear spring assumption.

2. **Interpolation Modeling Analysis**
   - **The client has also taken some careful lab data and would like to determine an interpolation relationship between the data points across the entire data range.** This carefully obtained lab data is stored in `labtest.mat`. Please use an appropriate interpolating technique to build this model. Justify the use of this interpolation technique and compare it to the performance of the regression analysis by discussing relevant fitting metrics and any salient features of each model that you deem appropriate.

3. **Extrapolation Analysis**
   - **The client is interested in predicting what the spring forces will look like if they extend the range of compression for the spring out to 0.4m.** Comment on how your models from your regression and interpolation analysis would handle this extrapolation and determine which model is best suited for this task, as well as the predicted force values for the best model.

Deliverables:

- A detailed PowerPoint presentation addressing the tasks above that adheres to client requirements and the Burdell Inc. style guide. See the provided template for reference and **remember to focus primarily on delivering only data and analyses that are actionable for the client's requests**. The slides must at least include:
  - **Figures/data showing the analyses from task 1**, including but not limited to the fitting data, the results from each regression model candidate, and evidence that the model appropriately describes the data. Do not forget to report your goodness-of-fit metrics in addition to your candidate best model.

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- **Figures/data showing the analysis from task 2**, including but not limited to your comparisons between your interpolation model and your best regression candidate.
- **Figures/data showing the analysis from task 3**, including but not limited to your extrapolation results extending the range of motion to the larger parameter value. Do not forget to comment on which model appears to be a better predictor.

- A .zip file containing your MATLAB scripts and supporting functions that produces all figures and calculations contained in your slides and adhering to the coding guidelines at Burdell Inc. Please make sure that your .zip file contains the following code elements:
  - A script file named `springAnalysis_[last name].m` that calls all necessary subfunctions and produces all plots used in your presentation document, as well as produces all your numerical results.
  - A function `[ y_fit, residuals ] = lsRegression( x_data, y_data, 'fitType' )` which takes as input the *x* and *y* data for your regression fit as well as a string label for your choice of regression fit and returns the predicted fit values, `y_fit`, and the residuals between the fitted data and the observed data. When selecting fit type using your string label, it's intended that your `lsRegression` function has each regression analysis built into it and this string tells the function which code to use.
  - A function `[ rmseVal ] = rmsError( y_data, y_fit )` which takes as input the fitted and observed *y* data and outputs the root mean squared error, `rmseVal`.
  - Please be sure to check case and naming for all functions – they will be used in a larger coding environment with many contributors and any deviations from the above formatting will lead to errors. This will significantly impact my evaluation of your performance.The script must run in a fresh environment/directory with no variables pre-loaded into the workspace and must return the requested values and figures.

I want you to determine the number of slides and the content of those slides that will make the most compelling case for your conclusions based on the client's needs. Please ensure that your deliverables are clear and concise, as I will be presenting them to the client. Because I haven't performed this analysis myself, I need you to provide detailed presenter notes that explain all your modeling choices, including but not limited to the context under which you are using your numerical frameworks, any relevant information for the client about how they work, and how you analyze the appropriateness and trustworthiness of the results. If you have any questions or need further clarification, do not hesitate to reach out to me.

I look forward to your insightful analysis and contributions to this project.

Best regards,
Anthony Jordan
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